

MELODY YIYUAN ZHANG

Ph.D. Candidate, University of Michigan, Ann Arbor, MI 48105

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EDUCATION

Ph.D. in Chemical Engineering and Scientific Computing University of Michigan, Ann Arbor, MI	<i>Expected May 2027</i> GPA: 3.81
B.S. in Chemical Engineering Purdue University, West Lafayette, IN	<i>May 2022</i> GPA: 3.82

AWARDS

Empowering Research with AI Award, AI Institutes at Michigan (AIIM)	2026
Best Overall Poster Award, U-M Annual Data Science & AI Summit	2025
Michigan Institute for Computational Discovery & Engineering (MICDE) Fellowship	2025
2nd Place, U Michigan Ross Business+Tech Datathon	2025
1st Place, College of Engineering Undergraduate Research Conference	2021
Research Grant, Office of Undergraduate Research, Purdue University	2021
Purdue Women in Engineering Program Scholarship	2021
Chemical Engineering Stephen Craig Outstanding Junior Award	2021
AstraZeneca Rising Star Award for initiative, innovation, and productivity	2021

RESEARCH EXPERIENCE

Graduate Student Research Assistant , Lindsey Group & Glotzer Group	<i>Oct. 2022 – Present</i>
- Extended the machine-learned interaction model (ML-IAM), ChIMES, to describe how effective interactions between polymer-grafted nanoparticles depend on key physicochemical attributes. - Integrated the extended ChIMES model with digital alchemy inverse design for efficient optimization of particle attributes toward target self-assembled structures. - Designed a generalized ML-IAM framework that models anisotropic interactions for anisotropic nanoparticles with convex shapes, enabling broad applicability to nanomaterial systems. - Developed an adaptive parallel meshing algorithm for training data generation that reduces required data volume by $\sim 100\times$ by focusing on key features of the target potential energy surface. - Contributed to the development and improvement of state-of-the-art machine-learning and simulation software, including ChIMES, HOOMD-blue, and freud.	
Undergraduate Research Assistant , Dou Group, Purdue University	<i>Jan. 2020 – May 2022</i>
- Modified the surface of formamidinium-based perovskites using synthesized ligands to passivate defects and kinetically trap the photovoltaic crystal phase. - Incorporated amine-thiol chemistry in the perovskite layer and achieved a power conversion efficiency (PCE) of 20.1% and improved stability under ambient conditions. - Streamlined the data analysis process of photovoltaic device characteristics and performance.	

RESEARCH PUBLICATIONS

Zhang, M. Y., Lee, S. K. A., Glotzer, S. C., and Lindsey, R. K. A Generalized Machine-Learning Interaction Model for Anisotropic Interactions in Coarse-Grained Nanoparticles. Manuscript in preparation.

Zhang, M. Y.*, Lee, S. K. A.*, Glotzer, S. C., and Lindsey, R. K. A Generalized Machine-Learning Framework for Developing Alchemical Many-Body Interaction Models for Polymer-Grafted Nanoparticles. *J. Chem. Theory Comput.* 2025, 21, 9853–9867. DOI: [10.1021/acs.jctc.5c00901](https://doi.org/10.1021/acs.jctc.5c00901) (*equal contribution)

Finkenauer, B., **Zhang, Y.**, Ma, K., Turnley, J. W., Schulz, J., et al. Amine-Thiol/Selenol Chemistry for Efficient and Stable Perovskite Solar Cells. *J. Phys. Chem. C* 2023, 127, 930–938. DOI: [10.1021/acs.jpcc.2c06967](https://doi.org/10.1021/acs.jpcc.2c06967)

Zhang, Y. Interface Engineering to Kinetically Trap the Photovoltaic Crystal Phase of Formamidinium Lead Iodide Perovskite. *Journal of Purdue Undergraduate Research* 2021, 11, Article 15. DOI: [10.7771/2158-4052.1533](https://doi.org/10.7771/2158-4052.1533)

CONFERENCE PRESENTATIONS

Zhang, M. Y., Lee, S. K. A., Glotzer, S. C., and Lindsey, R. K. A Generalized Alchemical Machine-Learned Interaction Potential for Modeling Anisotropic Nanomaterial Interactions. Oral presentation, *Material Research Society*, Spring 2026, Honolulu, HI.

Zhang, M. Y., Lee, S. K. A., Glotzer, S. C., and Lindsey, R. K. A Generalized Framework for Developing Alchemical Coarse-Grained Interaction Models for Self-Assembling Polymer-Grafted Nanoparticles. Oral presentation, *Midwest Thermodynamics and Statistical Mechanics Conference*, Summer 2025, Madison, WI.

Zhang, M. Y., Lee, S. K. A., Laubach, B., Glotzer, S. C., and Lindsey, R. K. Generating Nanomaterial Design Surfaces with Physics-Informed Machine Learning Models. Oral presentation, *American Institute of Chemical Engineers*, Fall 2024, San Diego, CA.

Zhang, M. Y., Lee, S. K. A., Glotzer, S. C., and Lindsey, R. K. Towards Mapping Nanomaterial Design Space with Physics-Informed Machine-Learned Models. Poster presentation, *Midwest Thermodynamics and Statistical Mechanics Conference*, Summer 2024, Ann Arbor, MI.

Zhang, M. Y., Lee, S. K. A., Laubach, B., Tsai, S., Anderson, J. A., Glotzer, S. C., and Lindsey, R. K. A Machine-Learning Strategy for the Physics-Informed Design of Nanomaterials Using ChIMES. Poster presentation, *American Physical Society*, Spring 2024, Minneapolis, MN.

PROFESSIONAL EXPERIENCE

Advanced Modeling and AI Solutions Intern, Micron Technology, Boise, ID *May – Aug. 2026*

- Designed a material search agent that unifies naming conventions across multiple databases and intelligently interprets non-canonical user queries.
- Integrated automated data retrieval with process and structural modeling software, accelerating simulation setup and reducing manual engineering effort.

Validation Engineering Intern, AstraZeneca, Mt. Vernon, IN *May – July 2021*

- Employed the BIOVIA Discoverant data analysis software to compile an annual product quality report on two extensively produced active pharmaceutical ingredients in the company.
- Designed a computer application using Tulip to provide intuitive instructions for equipment and process risk assessments for the engineering and operation teams.

Resident Assistant, Purdue University *Nov. 2020 – May 2022*

LEADERSHIP EXPERIENCE

Liaison to the Director, Center for Simulation and Data Intensive Research in Chemical Engineering (CSiDIR-CHE), University of Michigan *2023 – 2024*

Engineering Peer Mentor, Women in Engineering Program, Purdue University *2020*

Engineering Peer Mentor, Boiler Student Network, Purdue University *2020*

Events Planning Chair, UR Multicultural Connections, Purdue University *2020 – 2021*

Purdue AIChE Officer, Student Relations & Faculty Relations Committee *2020*

Vice President, Wiley Resident Hall Club, Purdue University *Aug. 2019 – 2020*

RELEVANT COURSEWORK

Design and Analysis of Algorithms	Simulation of Condensed Matter
Machine Learning	Statistical & Irreversible Thermodynamics
Parallel Computing	Applied Math for Chemical Engineering
Numerical Linear Algebra	Big Data Management Tools

TECHNICAL SKILLS

Programming & HPC: Python, C++, Bash, Git, MPI, OpenMP, CUDA

Machine Learning: Variational Autoencoders, SBERT, Graph Neural Networks, scikit-learn, PyTorch

Molecular Simulation & Visualization: LAMMPS, HOOMD-blue, OVITO, VMD, Blender